T-11

$H_0: \xi \sim p_0(x) = \{(0,1)\}$

$H_1: \xi \sim p_1(x) = \frac{e}{e-1} e^{-x} \{(0,1)\}$

a) $n = 1$

$$\frac{p_1(x)}{p_0(x)} = \frac{e}{e-1} e^{-x} \geq C, \text{ где } C = const$$

$e^{-x} \geq B$

$x \leq A(G)$

$$p(x \leq A | H_0) = \int_0^A dx = A$$

$G: x \leq \alpha$

$\alpha_1 = \alpha$

$$p(x \leq A | H_1) = \int_0^\alpha \frac{e}{e-1} e^{-x} dx = \frac{e}{e-1}(1 - e^{-\alpha}).$$

$$\alpha_2 = 1 - \frac{e}{e-1}(1 - e^{-\alpha})$$

б) $$\frac{p_1(x_1) p_1(x_2)}{p_0(x_1) p_0(x_2)} = \left(\frac{e}{e-1}\right)^2 e^{-(x_1+x_2)} \geq C$$

$e^{-(x_1+x_2)} \geq B$

$x_1 + x_2 \leq A(G)$

$P(x_1+x_2 \le A \mid H_0) = \alpha$

$$\iint\limits_{x_1+x_2\le A} dx_1\,dx_2 = \frac{A^2}{2} \le \alpha$$

$A = \sqrt{2\alpha}$

$\delta: x_1 + x_2 \le \sqrt{2\alpha}$

$\alpha_1 = \alpha$

$$P(x_1+x_2 \le A \mid H_1) = \left(\frac{e}{e-1}\right)^2 \iint\limits_{x_1+x_2\le A} e^{-x_1} e^{-x_2}\,dx_1\,dx_2$$

$$= \int_0^A e^{-x_1}\left(1-e^{-(A-x_1)}\right)dx_1 = 1-e^{-A}-Ae^{-A}$$

$$1-\left(\frac{e}{e-1}\right)^2\left(1-e^{-\sqrt{2\alpha}}-\sqrt{2\alpha}\,e^{-\sqrt{2\alpha}}\right) = \alpha_2$$

$$\frac{L_1}{L_0} = \prod_{i=1}^{n}\frac{p_1(x_i)}{p_0(x_i)} \ge c$$

$$\ln\left(\frac{L_1}{L_0}\right) = \sum_{i=1}^{n}\underbrace{\ln\left(\frac{p_1(x_i)}{p_0(x_i)}\right)}_{\ln p_1(x_i)-\ln p_0(x_i)} \ge \ln c$$

$\sum \ln\left(\frac{p_1(x_i)}{p_0(x_i)}\right) \ge \ln c$

$$\ln\left(\frac{p_1(x_i)}{p_0(x_i)}\right) = \eta_i$$

$$\frac{\sum \eta_i - nM_\eta}{\sqrt{nD\eta_1}} \sim N(0,1)$$

$$P\left(\ln\frac{L_1}{L_0} \geq \ln C \mid H_0\right) = \alpha$$

$$\eta = \ln\frac{e}{e-1} - x$$

$$\ln\frac{L_1}{L_0} = \sum_{i=1}^{n} \ln\frac{e}{e-1} - \sum_{i=1}^{n} x_i \geq \ln C$$

$$G: \sum_{i=1}^{n} x_i \leq A$$

$$P\left(\frac{\sum x_i - nM_x}{\sqrt{nD_x}} \leq \frac{A - nM_x}{\sqrt{nD_x}} \mid H_0\right) = \alpha$$

норм распр:

$$\frac{A - n\cdot\frac{1}{2}}{\frac{1}{2\sqrt{3}}\sqrt{n}} = u_\alpha$$

$$A = \frac{n}{2} + \frac{\sqrt{n}}{2\sqrt{3}} u_\alpha$$

$$G: \sum_{\alpha_1 = \alpha} x_i \leq n\cdot\frac{1}{2} + \frac{\sqrt{n}}{2\sqrt{3}} u_\alpha$$

$$P\left(\sum x_i \leq A \mid H_1\right) = P\left(\frac{\sum x_i - nM_x}{\sqrt{nD_x}} \leq \frac{A - nM_x}{\sqrt{nD_x}} \mid H_1\right)$$

Найдем Mx и Mx^2

$$Mx = \int_0^1 x \frac{e}{e-1} e^{-x} dx = \frac{-e}{e-1}\left(xe^{-x}\Big|_0^1 + \int_0^1 e^{-x} dx\right)$$

$$= \frac{e-2}{e-1}$$

$$Mx^2 = \int_0^1 x^2 \frac{e}{e-1} e^{-x} dx = \frac{2e-5}{e-1}$$

$$Dx = Mx^2 - (Mx)^2 = \frac{e^2-3e+1}{(e-1)^2}$$

$$P\left(\frac{\sum x_i - nMx}{\sqrt{nDx}} \le \frac{A - nMx}{\sqrt{nDx}} \,|\, H_1\right) = \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$$\gamma = \frac{u_\alpha \cdot \frac{\sqrt{n}}{2\sqrt{3}} - n\frac{e-2}{e-1} + \frac{1}{2}n}{\sqrt{\frac{n(e^2-3e+1)}{e-1}}}$$

$$\alpha_2 = 1 - \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$\gamma \xrightarrow[n\to\infty]{} 1 \Rightarrow$ критерий состоятел.

Д: 6: $x_{min} \le c$

$$P(x_{min} \le c) = 1 - (1 - F(c))^n = \alpha$$

$$c = 1 - \sqrt[n]{1-\alpha}$$

$$\alpha = \alpha_1$$

$P(X_{min} \le c | H_1) = \alpha$

$$W = \left(1 - \left(1 - \frac{e}{e-1}\left(1 - e^{-1 + \sqrt[n]{1-\alpha}}\right)\right)^n \xrightarrow[n\to\infty]{} 1$$

$\Leftarrow$

$$W = \left(1 - \left(1 + e\frac{\ln(1-\alpha)}{e-1}\right) \cdot \frac{1}{n} + O\left(\frac{1}{n}\right)\right)^n \xrightarrow[n\to\infty]{} 1$$

$\Rightarrow$ ее явл. сост.